Introduction to the Laws of Motion

Lesson 5: Newton's Second Law

Newton's Second Law – Study Notes

Key Concepts

1. **Force–Mass–Acceleration Relationship** – The net external force acting on an object equals the product of its mass and its acceleration: **\(\mathbf{F}=m\mathbf{a} \)**.

2. **Mass as Inertia** – Mass measures an object’s resistance to changes in its state of motion (its inertia).

3. **Vector Nature** – Force and acceleration are vectors; they have both magnitude *and* direction.

4. **Net (Resultant) Force** – Only the vector sum of all forces determines the acceleration.

5. **Units** – In the SI system, force is measured in newtons (N), mass in kilograms (kg), and acceleration in meters per second squared (m /s {^{2}).

Summary

Newton’s Second Law provides the quantitative link between how much force you apply to an object and how quickly its velocity changes. The law states that the **net external force** on an object is equal to the **mass** of the object multiplied by its **acceleration** (\(F = ma\)). Because both force and acceleration are vectors, the direction of the acceleration is the same as the direction of the net force.

Mass is a measure of an object’s inertia—its tendency to resist changes in motion. A larger mass requires a larger force to achieve the same acceleration as a smaller mass. Conversely, for a given force, a lighter object will accelerate more quickly. This principle underlies everything from everyday activities (pushing a shopping cart) to engineering design (calculating thrust for rockets).

In practice, you often rearrange the formula to solve for the unknown quantity:

- **Find acceleration:** \(a = \dfrac{F}{m} \)
- **Find force:** \(F = m \times a \)
- **Find mass:** \(m = \dfrac{F}{a} \)

Remember to keep track of units and vector directions when solving problems.

Important Points

- **Net force** = sum of all individual forces (vector addition).
- **Acceleration** is the rate of change of velocity (\(\Delta v / \Delta t \)).
- **1 newton (N)** = the force needed to accelerate **1 /kg** of mass at **1 /m /s {^{2}**.
- If the net force is zero, acceleration is zero !’ the object moves at constant velocity (including rest).
- The law applies in inertial reference frames (non accelerating frames).

Examples

Example 1 – Calculating Acceleration

A crate with a mass of **15 /kg** is pulled across a floor by a horizontal force of **45 /N**.

Solution:

\[

a = \frac{F}{m} = \frac{45\ \text{N}}{15\ \text{kg}} = 3\ \text{m /s}^{-2}

\]

The crate accelerates at **3 /m /s {^{2}** in the direction of the applied force.

Example 2 – Determining Required Force

A 2 kg hockey puck must be accelerated from rest to a speed of **6 /m /s {^{1}** in **0.5 /s**. What constant force is needed?

1. Find acceleration:

\[

a = \frac{\Delta v}{\Delta t}= \frac{6\ \text{m /s}^{-1}}{0.5\ \text{s}} = 12\ \text{m /s}^{-2}

\]

2. Apply \(F = ma\):

\[

F = 2\ \text{kg} \times 12\ \text{m /s}^{-2}= 24\ \text{N}

\]

A constant force of **24 /N** in the direction of motion will produce the required acceleration.

Quick Review

1. What does the equation \(F = ma\) tell you about the relationship between force, mass, and acceleration?

2. If a 10 /kg object experiences a net force of 50 /N, what is its acceleration?

3. Why must you consider the *net* force rather than a single applied force?

4. How would the acceleration change if the mass were doubled while the applied force stayed the same?

5. State the SI unit for force and explain how it is derived from the base units.

Further Reading

• **Newton’s First Law (Inertia) and Third Law (Action–Reaction)** – how the three laws work together.

• **Free body diagrams** – visual tools for finding net forces.

• **Friction and air resistance** – forces that often affect the net force.

• **Dynamics in non inertial frames** – pseudo forces and apparent acceleration.

• **Work, Energy, and Power** – connecting force and motion to energy concepts.